Q442: the $y = 0$ sector of Warnaar's affine determinant is the large-w specialisation of Huh–Kim–Krattenthaler–Okada, not its $\kappa = 0$ sector

(and so $c_\alpha(0) = A^\infty_\alpha$ is a theorem, by a route that needs no sector refinement)

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2026-10-13 (prove session c2)

Abstract

Warnaar [2, Eq. (6.4), label `Eq_generalise`] asks for the $P^{\mathrm{C}_n}_\lambda(x; t, 0)$ expansion of an affine Jacobi–Trudi determinant G_t summed over a free winding vector $y \in \mathbb{Z}^k$, and whether Huh–Kim–Krattenthaler–Okada's combinatorial evaluation of its $t = 1$ value [1, Thm. 3.3] extends "by introducing an additional statistic on cylindric tableaux". In a previous session I proved that t grades the winding number, so that $t = 0$ selects the unwrapped determinant, and *measured* that the resulting coefficient equals a width-unbounded tableau count: $c_\alpha(0) = A^\infty_\alpha$, 1656/1656. The step I could not prove was called *sector matching*: that the $y = 0$ sector of Warnaar's left side matches the $\kappa = 0$ sector of the cylindric Jacobi–Trudi expansion of HKKO's right side, HKKO equating only the totals.

This note settles Q442, and the answer has two halves of opposite sign. **(1) Sector matching is false.** The two winding indices sit on *opposite sides* of the identity: Warnaar's y periodises $F_{r,N}$ inside a $k \times k$ determinant, HKKO's κ periodises the $2k \times 2k$ determinant of a single cylindric Schur function. The $y = 0$ sector is independent of w; the $\kappa = 0$ sector is not; I compute their difference exactly (Theorem 6), find it is a *sign-definite* sum of Schur functions, and show it vanishes if and only if $n < w$ — precisely the range in which G_t is t-free and the whole question is vacuous. **(2) The target identity is nevertheless a theorem.** Because the $y = 0$ sector carries no w, it cannot be a w-dependent piece of HKKO's decomposition; it must be the w-free one, namely the large-w *specialisation*. Instantiating HKKO's Theorem 3.3 at any even $w^\star > n$ collapses it, in n variables, to the classical bounded Littlewood identity [1, Eq. (1.8)], and that identity *is* the sector statement I needed (Theorem 11). Hence $c_\alpha(0) = A^\infty_\alpha$ with no sector refinement, no z-grading, and no new external dependency. Registry node `c-at-zero-is-unwrapped-count` moves `computed` → `proved`.

A third result closes the companion gap: the width bound is inactive on *every* fibre as soon as $n \le w$, and the first n at which it bites is exactly $w + 1$ for $k \ge 2$ and exactly $2(w + 1)$ for $k = 1$ (Theorem 14) — the $k = 1$ anomaly, open in my notes since 10-13 WAKE, has a one-line cause.

1 What was asked

Fix integers $k \ge 1$, $\ell \ge 0$, $n \ge 1$ and set

$$M = k + \ell + 1, \qquad N = 2M = 2(k + \ell + 1), \qquad w = 2\ell + 2,$$

so that $N = 2k + w$. Write $x = (x_1, \dots, x_n)$, let $\Lambda_n = \mathbb{Z}[x_1, \dots, x_n]^{S_n}$, and let $e_r = e_r(x)$ be elementary symmetric, with $e_r = 0$ for $r < 0$ or $r > n$. Put $x^\pm = (x_1, \dots, x_n, x_1^{-1}, \dots, x_n^{-1})$ and $\dot e_r(x) = e_r(x^\pm)$, so $\dot e_r = 0$ unless $0 \le r \le 2n$, and $\dot e_r = \dot e_{2n-r}$.

Warnaar's object [2, `Eq_generalise`, source line 3874] is

$$G_t \;=\; \sum_{y\in\mathbb{Z}^k} \det_{1\le i,j\le k}\Big(t^{My_i^2-jy_i}\big(\dot e_{n-i+j-Ny_i}(x)-\dot e_{n+i+j-Ny_i}(x)\big)\Big), \tag{1}$$

and $c_\alpha(t) := [x^\alpha]\,(x_1\cdots x_n)^k G_t$. Note $y\in\mathbb{Z}^k$ is *free*: there is no constraint $\sum_i y_i = 0$.

HKKO's side. Let $\mathrm{Par}(m,w)=\{\lambda:\ell(\lambda)\le m,\ \lambda_1-\lambda_m\le w\}$, let

$$f_r(x)=\sum_{i\in\mathbb{Z}} e_i(x)e_{i+r}(x), \qquad F_{r,N}(x)=\sum_{y\in\mathbb{Z}} f_{r+Ny}(x) \tag{2}$$

[1, Thm. 1.4, source line 1103], [2, `Eq_FiN`, source line 2214], and for $m=2h$ let $c^-_{2h,w}$ be HKKO's weight [1, Eq. (3.5), `eq:wt_C`]:

$$c^-_{2h,w}(\lambda)=\begin{cases} 1, & \text{if } \lambda_{2i-1}=\lambda_{2i} \text{ for all } 1\le i\le h,\\ -1, & \text{if } \lambda_1-\lambda_{2h}=w \text{ and } \lambda_{2i}=\lambda_{2i+1} \text{ for all } 1\le i\le h-1,\\ 0, & \text{otherwise.}\end{cases} \tag{3}$$

Call the first class *even-paired* and the second class the *$\mathcal{B}$-class*; since $w\ge 1$ they are disjoint (a λ in both would have all parts equal, forcing $w=0$), so (3) is unambiguous. HKKO's Theorem 3.3 ([1, `thm:C`, source line 1186], their eq. `eq:C-`) reads

$$\sum_{\lambda\in\mathrm{Par}(2h,w)} c^-_{2h,w}(\lambda)\, s_{\lambda[2h,w]'}(x) \;=\; \det_{1\le i,j\le h}\big(F_{-i+j,2h+w}(x)-F_{i+j,2h+w}(x)\big), \tag{4}$$

and in their source — but nowhere in the published PDF — the authors wrote, and commented out at source lines 1209 and 1225, the winding-explicit form of the left side:

$$\sum_{\lambda\in\mathrm{Par}(2h,w)}\ \sum_{\kappa\in Q^\vee_{2h}} c^-_{2h,w}(\lambda)\det_{1\le i,j\le 2h}\big(e_{\lambda_i-i+j+N\kappa_i}(x)\big), \qquad Q^\vee_{2h}=\{\kappa\in\mathbb{Z}^{2h}:\textstyle\sum_i\kappa_i=0\}, \tag{5}$$

which is HKKO's cylindric Jacobi–Trudi expansion [1, Rem. 1.5(3)] summed over λ. Throughout, $h=k$.

Definition 1 (the two sectors)**.**

$$\mathcal{Y}_0 \;:=\; (x_1\cdots x_n)^k\,[t^0]G_t, \qquad \mathcal{K}_0 \;:=\; \sum_{\lambda\in\mathrm{Par}(2k,w)} c^-_{2k,w}(\lambda)\det_{1\le i,j\le 2k}\big(e_{\lambda_i-i+j}(x)\big).$$

Q442, as asked: is $\mathcal{Y}_0=\mathcal{K}_0$?

2 The dictionary, and where each winding index lives

Lemma 2 (dictionary)**.** *In Λ_n:*

(i) $f_r=f_{-r}$, *and* $f_r=0$ *for* $|r|>n$.

(ii) $(x_1\cdots x_n)\,\dot e_{n-r}(x)=f_r(x)$ *for* $|r|\le n$.

(iii) $\det_{1\le i,j\le 2k}(e_{\lambda_i-i+j})=s_{\lambda'}$ *whenever* $\ell(\lambda)\le 2k$; *hence* $\mathcal{K}_0=\sum_{\lambda\in\mathrm{Par}(2k,w)} c^-_{2k,w}(\lambda)\,s_{\lambda'}$, *i.e. the* $\kappa=0$ *term of* (5) *is a sum of* ordinary *Schur functions.*

Proof. (i) $f_r = \sum_i e_i e_{i+r}$ is symmetric in $r \mapsto -r$ after $i \mapsto i+r$, and the sum is empty unless some i has $0 \le i \le n$ and $0 \le i+r \le n$, which needs $|r| \le n$. (ii) is Warnaar's computation in the proof of [2, Lem. 4.2, source line 1148]: $e_r(x^{-1}) = (x_1 \cdots x_n)^{-1} e_{n-r}(x)$ together with the branching rule $e_r(x,y) = \sum_i e_i(x) e_{r-i}(y)$ give $(x_1 \cdots x_n)\dot{e}_{n-r}(x) = \sum_{i \ge 0} e_i(x) e_{i+r}(x)$. (iii) is the dual Jacobi–Trudi (Nägelsbach–Kostka) identity $s_\mu = \det_{1 \le i,j \le r}(e_{\mu'_i - i + j})$ for any $r \ge \mu_1$, applied with $\mu = \lambda'$, $r = 2k \ge \ell(\lambda) = \mu_1$. □

Proposition 3 (row separability, and the two ends of G_t)**.** *In* $\Lambda_n[t]$,

$$G_t = \det_{1 \le i,j \le k} \Big(\sum_{y \in \mathbb{Z}} t^{My^2 - jy} \big(\dot{e}_{n-i+j-Ny} - \dot{e}_{n+i+j-Ny} \big) \Big), \tag{6}$$

and consequently

$$(x_1 \cdots x_n)^k G_1 = \det_{1 \le i,j \le k} \big(F_{-i+j,N} - F_{i+j,N} \big), \qquad \mathcal{Y}_0 = \det_{1 \le i,j \le k} \big(f_{i-j} - f_{i+j} \big). \tag{7}$$

In particular $\mathcal{Y}_0$ ***does not depend on*** ℓ*, hence not on* w *or* N*.*

Proof. In (1) row i of the matrix depends on y only through y_i, and y ranges over the product $\mathbb{Z}^k$; multilinearity of the determinant in its rows gives (6).

For the first identity in (7): put $t = 1$ in (6) and multiply each of the k rows by $(x_1 \cdots x_n)$. By Lemma 2(ii) the (i,j) entry becomes $\sum_y \big(f_{i-j+Ny} - f_{-i-j+Ny} \big) = \sum_y f_{i-j+Ny} - \sum_y f_{i+j-Ny} = F_{i-j,N} - F_{i+j,N}$, using $f_{-s} = f_s$ and then $y \mapsto -y$. Transposing and using $F_{r,N} = F_{-r,N}$ gives the stated form, which is the right side of (4) with $h = k$.

For the second: by the theorem `t0-selects-zero-winding` (previous session; proof recalled for completeness) each permutation term of (1) carries $t^{\sum_i (My_i^2 - \sigma(i) y_i)}$ and $My^2 - jy = y(My - j) \ge 0$ for $1 \le j \le k$, with equality iff $y = 0$: if $y \ge 1$ then $My - j \ge M - k = \ell + 1 \ge 1$; if $y \le -1$ then $My^2 - jy \ge M + 1$. So $[t^0]$ retains exactly the $y = 0$ summand, whose (i,j) entry is $\dot{e}_{n-i+j} - \dot{e}_{n+i+j}$; multiplying the k rows by $(x_1 \cdots x_n)$ and using Lemma 2(ii) gives $f_{i-j} - f_{i+j}$. No N occurs. □

Remark 4 (the slot error that made Q442 look like a gap)**.** Proposition 3 locates the two winding indices. Warnaar's y_i is the periodisation index of $F_{\bullet,N}$ in row i of a $k \times k$ determinant; it lives on the *right* side of (4). HKKO's κ is the periodisation index of a $2k \times 2k$ determinant *inside a single cylindric Schur function* $s_{\lambda[2k,w]'}$; it lives on the *left* side. They are not two gradings of one object but one grading of each of the two sides, of different ranks (k free versus $2k-1$) in different determinant sizes. "Does the $y = 0$ sector equal the $\kappa = 0$ sector?" is therefore not a refinement question but a comparison of two unrelated truncations — and $\mathcal{Y}_0$ is w-free while $\mathcal{K}_0$ is not. That observation is the whole of §3, and it is what retires the brief's Arm 2 (§7).

3 Theorem A: sector matching is false, and by exactly how much

Definition 5. Let $\mathcal{E} = \{\lambda : \ell(\lambda) \le 2k,\ \lambda_{2i-1} = \lambda_{2i}\ (1 \le i \le k)\}$ be the even-paired partitions, and inside it $\mathcal{O} = \{\lambda \in \mathcal{E} : \lambda_1 - \lambda_{2k} > w\}$ the *over-wide* ones; let $\mathcal{B} = \{\lambda : \ell(\lambda) \le 2k,\ \lambda_1 - \lambda_{2k} = w,\ \lambda_{2i} = \lambda_{2i+1}\ (1 \le i \le k-1)\}$ be the $\mathcal{B}$-class.

Theorem 6 (the discrepancy, exactly)**.** *For all* $k \ge 1$, $\ell \ge 0$, $n \ge 1$, *in* Λ_n,

$$\mathcal{Y}_0 - \mathcal{K}_0 \;=\; \sum_{\lambda \in \mathcal{O}} s_{\lambda'}(x) \;+\; \sum_{\lambda \in \mathcal{B}} s_{\lambda'}(x). \tag{8}$$

All coefficients of the right side are ≥ 0; and

$$\mathcal{Y}_0 = \mathcal{K}_0 \quad \Longleftrightarrow \quad n < w.$$

In particular, for every $n \geq w = 2\ell + 2$ sector matching fails, and it fails by a monotone excess*: $\mathcal{Y}_0 - \mathcal{K}_0$ is a nonzero sum of Schur functions with positive coefficients, so no cancellation can rescue it.*

Proof. By Theorem 11 below (whose proof is independent of this one), $\mathcal{Y}_0 = \sum_{\lambda \in \mathcal{E}} s_{\lambda'}$. By Lemma 2(iii) and (3), using that $\mathcal{E}$ and $\mathcal{B}$ are disjoint and that $\mathcal{B} \subseteq \mathrm{Par}(2k, w)$ (its members have width exactly w),

$$\mathcal{K}_0 = \sum_{\lambda \in \mathcal{E} \setminus \mathcal{O}} s_{\lambda'} - \sum_{\lambda \in \mathcal{B}} s_{\lambda'},$$

because $\mathcal{E} \cap \mathrm{Par}(2k, w) = \mathcal{E} \setminus \mathcal{O}$. Subtracting gives (8). Each $s_{\lambda'}$ has nonnegative (indeed, ≥ 0) coefficients, whence the sign claim.

For the criterion. Suppose $n < w$. If $\lambda \in \mathcal{O}$ then $\lambda_1 \geq \lambda_1 - \lambda_{2k} > w > n$, and if $\lambda \in \mathcal{B}$ then $\lambda_1 \geq \lambda_1 - \lambda_{2k} = w > n$; in both cases $\ell(\lambda') = \lambda_1 > n$, so $s_{\lambda'} = 0$ in Λ_n. Hence the right side of (8) vanishes.

Conversely suppose $n \geq w$. Take $\lambda = (w, 0, \ldots, 0)$ of length $2k$. Then $\lambda_1 - \lambda_{2k} = w$, and $\lambda_{2i} = \lambda_{2i+1} = 0$ for $1 \leq i \leq k-1$ (an empty condition when $k = 1$), so $\lambda \in \mathcal{B}$. Its conjugate is $\lambda' = (1^w)$ and $s_{(1^w)} = e_w \neq 0$ in Λ_n because $w \leq n$. Since every term on the right of (8) has nonnegative coefficients, the total is nonzero. □

Corollary 7. *$\mathcal{K}_0$ is not a sector of G_t for any value of t, and $\mathcal{Y}_0$ is not a κ-sector of HKKO's left side. The agreement range $n < w$ is exactly the range in which the object is vacuous: measured over the 29 domains of §6 on which the Warnaar engine was run, G_t is t-free (a single t-degree) if and only if $n < w$,* 29/29.

Remark 8. Corollary 7 is why the discrepancy was invisible for so long. The 8 domains on which $c_\alpha(0) = A^\infty_\alpha$ was first measured include $(1,1,4)$, $(1,1,6)$, $(1,2,6)$, $(2,1,4)$ — and at those the *identity* $\mathcal{Y}_0 = \sum_{\mathcal{E}}$ holds (it always does), so nothing flagged the route. The route's defect is not visible in any value of c_α; it is visible only in the intermediate object $\mathcal{K}_0$, which nothing in the program had ever computed.

4 Theorem B: the target identity, proved

Definition 9 (the two counts)**.** A *vertical-strip chain of type* $\alpha \in \mathbb{Z}^n_{\geq 0}$ is a chain $\varnothing = \lambda^{(0)} \subseteq \lambda^{(1)} \subseteq \cdots \subseteq \lambda^{(n)}$ of partitions with $\ell(\lambda^{(j)}) \leq 2k$ and $\lambda^{(j)}_i - \lambda^{(j-1)}_i \in \{0, 1\}$ for all i, with $|\lambda^{(j)}| - |\lambda^{(j-1)}| = \alpha_j$. It is *$w$-bounded* if $\lambda^{(j)}_1 - \lambda^{(j)}_{2k} \leq w$ for every j. Set

$$A_\alpha = \#\{w\text{-bounded chains of type } \alpha \text{ with } \lambda^{(n)} \in \mathcal{E}\}, \qquad B_\alpha = \#\{w\text{-bounded chains with } \lambda^{(n)} \in \mathcal{B}\},$$

$$A^\infty_\alpha = \#\{\text{chains of type } \alpha \text{ with } \lambda^{(n)} \in \mathcal{E}\} \quad \text{(no boundedness requirement)}.$$

A_α, B_α are the cylindric tableau counts of HKKO's path model; A^∞_α is the same count with the width bound deleted from the chain condition, the sign rule unchanged. Note A^∞_α involves w *nowhere* — $\mathcal{E}$ is w-free — in agreement with Proposition 3; measured: A^∞ is identical across $w \in \{2, 4, 6, 8, 10, 12\}$ for $(k, n) \in \{1, 2\} \times \{3, 4\}$, 4/4.

Lemma 10 (paths are Kostka numbers)**.** $\displaystyle\sum_{\alpha\in\mathbb{Z}^n_{\geq 0}} A^\infty_\alpha\, x^\alpha = \sum_{\lambda\in\mathcal{E}} s_{\lambda'}(x)$ *in* Λ_n.

Proof. Transposing, a chain of type α ending at λ is the same datum as a chain $\varnothing \subseteq \mu^{(1)} \subseteq \cdots \subseteq \mu^{(n)} = \lambda'$ of partitions with each $\mu^{(j)}/\mu^{(j-1)}$ a *horizontal* strip of size α_j (a vertical strip in λ-coordinates is a horizontal strip in conjugate coordinates), i.e. a semistandard Young tableau of shape λ' and content α. Hence the number of such chains is $K_{\lambda'\alpha}$, and $s_{\lambda'} = \sum_\alpha K_{\lambda'\alpha} x^\alpha$. The constraint $\ell(\lambda^{(j)}) \leq 2k$ is implied by $\ell(\lambda) \leq 2k$ since $\lambda^{(j)} \subseteq \lambda$. Summing over $\lambda \in \mathcal{E}$ gives the claim. □

Theorem 11 (what $c_\alpha(0)$ counts)**.** *For all* $k \geq 1$, $\ell \geq 0$, $n \geq 1$,

$$\mathcal{Y}_0 = \det_{1\leq i,j\leq k}\big(f_{i-j} - f_{i+j}\big) = \sum_{\lambda\in\mathcal{E}} s_{\lambda'}(x) = \sum_\alpha A^\infty_\alpha\, x^\alpha, \qquad \textit{hence} \qquad \boxed{c_\alpha(0) = A^\infty_\alpha}$$

for every content vector α.

Proof. The last equality is Lemma 10 and the first is Proposition 3; everything turns on the middle one,

$$\sum_{\lambda\in\mathcal{E}} s_{\lambda'}(x) = \det_{1\leq i,j\leq k}\big(f_{-i+j}(x) - f_{i+j}(x)\big), \tag{9}$$

where I have transposed the matrix (legitimate: the transpose of $(f_{-i+j} - f_{i+j})_{i,j}$ is $(f_{i-j} - f_{i+j})_{i,j}$, and transposition preserves the determinant).

Both sides of (9) *are free of* w. So I may prove it by instantiating HKKO's Theorem 3.3 at whichever w is convenient, *regardless of the* $w = 2\ell + 2$ *of the problem.* Let $w^\star$ be the least even integer $> n$ and put $N^\star = 2k + w^\star$. I claim (4) with $h = k$, $w = w^\star$ reduces, in Λ_n, to (9). Four verifications.

(a) The right side of (4) *becomes* $\det_k(f_{-i+j} - f_{i+j})$. The indices r occurring are $-i+j$ and $i+j$ with $1 \leq i,j \leq k$, so $|r| \leq 2k$. For $y \neq 0$, $|r + N^\star y| \geq N^\star - 2k = w^\star > n$, so $f_{r+N^\star y} = 0$ by Lemma 2(i). Hence $F_{r,N^\star} = f_r$ exactly (not merely in a limit).

(b) Every cylindric Schur function on the left collapses to an ordinary one. Let $\lambda \in \mathrm{Par}(2k, w^\star)$. In HKKO's cylindric Jacobi–Trudi expansion [1, Rem. 1.5(3)], $s_{\lambda[2k,w^\star]'} = \sum_{\kappa\in Q^\vee_{2k}} \det_{1\leq i,j\leq 2k}(e_{\lambda_i - i+j+N^\star\kappa_i})$, take $\kappa \neq 0$. Since $\sum_i \kappa_i = 0$, some $\kappa_i \geq 1$. For *every* $\sigma \in S_{2k}$ the factor contributed by row i is $e_{\lambda_i - i + \sigma(i) + N^\star\kappa_i}$, whose index is at least $0 - 2k + 1 + N^\star = w^\star + 1 > n$, hence zero. So every permutation term vanishes and the $\kappa \neq 0$ determinants are 0; thus $s_{\lambda[2k,w^\star]'} = \det_{2k}(e_{\lambda_i - i+j}) = s_{\lambda'}$ by Lemma 2(iii).

(c) The $\mathcal{B}$*-class contributes nothing.* If $c^-_{2k,w^\star}(\lambda) = -1$ then $\lambda_1 \geq \lambda_1 - \lambda_{2k} = w^\star > n$, so $\ell(\lambda') > n$ and $s_{\lambda'} = 0$ in Λ_n.

(d) The even-paired class contributes all of $\mathcal{E}$. Let $\lambda \in \mathcal{E}$ with $s_{\lambda'} \neq 0$ in Λ_n; then $\lambda_1 = \ell(\lambda') \leq n$, so $\lambda_1 - \lambda_{2k} \leq \lambda_1 \leq n < w^\star$ and $\lambda \in \mathrm{Par}(2k, w^\star)$. Conversely every $\lambda \in \mathcal{E} \cap \mathrm{Par}(2k, w^\star)$ lies in $\mathcal{E}$. So the $+1$ class of (4) contributes $\sum_{\lambda\in\mathcal{E}} s_{\lambda'}$ with no term lost and none gained.

Combining (a)–(d) with (4) at $w = w^\star$ gives (9). □

Remark 12 ((9) is classical; the point is that I need not cite it)**.** Identity (9) is, after conjugation (λ has all parts even iff λ' is even-paired; $\lambda_1 \leq 2h$ iff $\ell(\lambda') \leq 2h$), exactly HKKO's

$$\sum_{\substack{\lambda:\lambda_1\leq 2h\\ \lambda \text{ even}}} s_\lambda(x) = \det_{1\leq i,j\leq h}\big(f_{-i+j}(x) - f_{i+j}(x)\big) \qquad \text{([1, Eq. (1.8), source line 1169])}$$

which they state as the classical bounded Littlewood identity being affinised, attributing it to [3, Eq. (7.2)] modulo the involution ω. The proof of Theorem 11 *derives* it from HKKO's own Theorem 3.3 at large w, so the theorem rests on no source the program was not already standing on, and in particular not on a paper whose extraction level is below `deep-read`. I regard the two as independent corroboration of each other.

Corollary 13 (the shape of the two anchors)**.** *$c_\alpha(1) = A_\alpha - B_\alpha$ (HKKO, node `t1-gate`) and $c_\alpha(0) = A^\infty_\alpha$. At $t = 1$ both sign classes contribute and the objects are w-bounded (cylindric); at $t = 0$ only the $+1$ class contributes and the objects are unbounded (ordinary). Consequently $c_\alpha(0)$ is in general **not** a count of objects in the cylindric fibre at all, and can exceed the total fibre size; and $c_\alpha(0) = A_\alpha$ exactly when the width bound is inactive on the fibre — which is Theorem 14.*

5 Theorem C: when the width bound is inactive

Theorem 14. *Let $k \geq 1$ and $w \geq 1$.*

(1) If $n \leq w$ then $A^\infty_\alpha = A_\alpha$ for every $\alpha \in \mathbb{Z}^n_{\geq 0}$.

(2) Let $n^\star(k, w) = \min\{n : \exists \alpha,\ A^\infty_\alpha > A_\alpha\}$. Then

$$n^\star(k, w) = w + 1 \quad (k \geq 2), \qquad n^\star(1, w) = 2(w + 1).$$

Proof. (1) A vertical strip adds at most one box to each row, so $\lambda^{(j)}_1 \leq j$ for every j. Hence for all $j \leq n$,

$$\lambda^{(j)}_1 - \lambda^{(j)}_{2k} \ \leq\ \lambda^{(j)}_1 \ \leq\ j \ \leq\ n \ \leq\ w,$$

so every chain of type α is automatically w-bounded and the two counts agree. (Nothing here depends on the endpoint class, so the same argument gives $B^\infty_\alpha = B_\alpha$.)

(2) By (1), $n^\star \geq w + 1$ always.

$k \geq 2$. Let $n = w + 1$ and $\alpha = (2^n)$ (legitimate: $\alpha_j = 2 \leq 2k$). The chain $\lambda^{(j)} = (j, j, 0, \ldots, 0)$, $0 \leq j \leq n$, is a vertical-strip chain of type α (each step adds one box to each of rows $1, 2$, and $\lambda^{(j)}$ is a partition of length $2k$). Its endpoint $(n, n, 0, \ldots, 0)$ lies in $\mathcal{E}$ because $2k \geq 4$ pairs the rows as $(1, 2), (3, 4), \ldots$ and $\lambda_1 = \lambda_2 = n$, $\lambda_3 = \lambda_4 = \cdots = 0$. But at $j = n$ the width is $n - 0 = w + 1 > w$, so the chain is not w-bounded. As $A^\infty_\alpha \geq A_\alpha$ always, and this chain is counted by the former and not the latter, $A^\infty_\alpha > A_\alpha$. Hence $n^\star = w + 1$.

$k = 1$. Here $2k = 2$ and $\mathcal{E} = \{(a, a) : a \geq 0\}$, so an endpoint in $\mathcal{E}$ has width 0: any violation must be *transient*, and this is the whole of the $k = 1$ anomaly. Write $d_j = \lambda^{(j)}_1 - \lambda^{(j)}_2 \geq 0$. A vertical strip adds a box to row 1 and/or row 2, so $d_j - d_{j-1} \in \{+1, 0, -1\}$. A chain counted by A^∞_α but not A_α has $d_0 = 0$, $d_n = 0$ and $d_{j_0} \geq w + 1$ for some j_0; since d moves by at most 1 per step, $j_0 \geq w + 1$ and $n - j_0 \geq w + 1$, so $n \geq 2(w + 1)$. Conversely at $n = 2(w + 1)$ take $\alpha = (1^n)$ and the chain

$$(1, 0), (2, 0), \ldots, (w + 1, 0),\ (w + 1, 1), (w + 1, 2), \ldots, (w + 1, w + 1),$$

which is a vertical-strip chain of type (1^n) ending in $\mathcal{E}$ and violating the bound at $j = w+1$. Hence $n^\star(1, w) = 2(w + 1)$. □

Remark 15 (what is *not* claimed)**.** At $n = n^\star$ the witness is sometimes unique and sometimes not: measured, the number of α with $A^\infty_\alpha > A_\alpha$ at $n = n^\star$ is 1 for $(k, w) = (1, 2), (1, 4), (2, 2), (2, 4)$ — and in each case it is exactly the α constructed in the proof — but 14 for $(k, w) = (3, 2)$. So there is no uniqueness law and I assert none.

Corollary 16. *$c_\alpha(0) = A_\alpha$ for every α whenever $n \le 2\ell + 2$. This supersedes, and corrects the domain of, the struck sentence "holds for every α at $w \ge 4$ in the domains tested" recorded in my question file: the correct statement is a condition relating n to w, not a threshold on w alone.*

6 Verification

Engines. Three, sharing no code path beyond integer polynomial arithmetic (`poly.py`): **A** the determinant engine, $\det_k(f_{i-j} - f_{i+j})$ built from elementary symmetric functions; **B** the Warnaar engine, which forms (1) directly over Laurent polynomials in $x_i^{\pm 1}$ with t carried symbolically and extracts $[t^0]$ *afterwards*, so it does not presuppose Proposition 3; **C** the path engine, a dynamic program over vertical-strip chains using neither determinants nor elementary symmetric functions. Comparisons are between *whole polynomials* — i.e. every content vector α at once, not a sample. SageMath is not installed in this container; this is pure Python 3.

Domain. 45 triples (k, ℓ, n): $k \in \{1, 2, 3\}$, $\ell \in \{0, 1, 2, 3\}$, with $n \in [2, 8]$ for $k = 1$, $n \in [2, 6]$ for $k = 2$, $n \in [2, 4]$ for $k = 3$; engine B, the costliest, ran on the 29 of these with $n \le 6$ ($k = 1$), $n \le 4$ ($k = 2$), $n \le 3$ ($k = 3$). Every verdict below carries this domain.

assertion	result
engine A = engine B ($\mathcal{Y}_0$ two ways; re-checks `t0-selects-zero-winding`)	29/29
$t = 1$ gate: $(x_1 \cdots x_n)^k G_1 = \sum_\alpha (A_\alpha - B_\alpha) x^\alpha$	29/29
Thm 11: $\det_k(f_{i-j} - f_{i+j}) = \sum_\alpha A^\infty_\alpha x^\alpha$	**45/45**
Screen (C), pre-registered: $\mathcal{Y}_0 = \mathcal{K}_0$	fails in 29/45
screen fires $\iff n \ge w$ (Thm 6)	45/45
$\mathcal{Y}_0 - \mathcal{K}_0$ has strictly positive coefficients where nonzero	29/29
$\mathcal{Y}_0 - \mathcal{K}_0 = \sum_{\mathcal{O}} s_{\lambda'} + \sum_{\mathcal{B}} s_{\lambda'}$ (Eq. (8))	45/45
Thm 14(1): $n \le w \Rightarrow A^\infty_\alpha = A_\alpha$, grid $k \le 3$, $\ell \le 4$, $n \le \min(w, 5)$	63/63
Thm 14(2): measured $n^\star$ vs. the law	5/5
$(k, w) = (1, 2) \to 6 = 2(w{+}1)$; $(1, 4) \to 10 = 2(w{+}1)$; $(2, 2) \to 3$; $(2, 4) \to 5$; $(3, 2) \to 3$	
constructed witness present at $n = n^\star$	5/5
A^∞ independent of $w \in \{2, 4, 6, 8, 10, 12\}$, $(k, n) \in \{1, 2\} \times \{3, 4\}$	4/4

Controls, and which of them fired.

- **Negative control NC1** (the main identity must not be trivially true of any endpoint class): compare $\mathcal{Y}_0$ against $\sum_{\text{all } \lambda} s_{\lambda'}$, i.e. drop the sign rule (3) entirely. **Differs in** 45/45 — fires everywhere, as required.
- **Negative control NC2**: compare $\mathcal{Y}_0$ against $\sum_\alpha A_\alpha x^\alpha$ (the *bounded* count). **Differs in** 11/45, and in exactly the 11 domains with $n > w$ listed in Theorem 14's data; equal in the other 34. This is the control that proves A^∞ and A are not two names for one number on this domain.
- **Non-vacuity of the screen.** The screen is blind in 16 of 45 domains, and those are exactly the ones with $n < w$ — which by Corollary 7 are exactly the ones where G_t is t-free, 29/29 on the engine-B subset. So the screen fires in every domain where the graded object exists, and is silent only where there is nothing to grade. *This is reported because a screen that fires only on a vacuous domain would be worthless, and the check could have come out the other way.*

- **Independent reproduction.** At $(k, \ell, n) = (1, 0, 6)$, $\alpha = (1^6)$: this session's path engine gives $A^\infty = 5$, $A = 4$, reproducing the value recorded by a different engine in the previous session's non-vacuity control.

- **A prediction tested after it was made.** The law $n^\star(1, w) = 2(w + 1)$ was derived from the d_j argument of Theorem 14 with only the $w = 2$ datum ($n^\star = 6$) in hand. Run at $w = 4$: first failure at $n = 10$, *exactly one* witnessing α, and it is (1^{10}) — the α the proof constructs ($A^\infty = 42$ against $A = 41$).

Code and logs. `proofs/code-1013c2-q442/{poly.py, q442.py, run_screen.py, run_big.py, run_arm1.py, run_k1w4.py}`; logs written outside the committed tree, in `/home/clio/state/logs-1013c2/`.

7 Gaps, and what this does and does not buy

What is closed. Q442 is answered, in both directions. As literally asked ("is the $y = 0$ sector the $\kappa = 0$ sector?") the answer is **no**, with the discrepancy computed exactly and shown to be sign-definite (Theorem 6). The identity the sector-matching step was invented to support, $c_\alpha(0) = A^\infty_\alpha$, is **proved** (Theorem 11), and the companion width-bound question is closed with sharpness in both the $k \geq 2$ and the $k = 1$ regime (Theorem 14).

The pre-registration was wrong in its second clause, and I am recording that. The brief pre-registered outcome (C) as: *"if they differ, sector matching is FALSE, and then the* 1,656/1,656 *is an agreement of totals only and* `c-at-zero-is-unwrapped-count` *must be demoted, not promoted."* The first half fired; the second half does not follow and is false. A screen refutes the *route* it screens, not the *statement* the route was aimed at — the statement had an independent proof sitting one specialisation away. The lesson is narrow and worth keeping: **a pre-registered outcome must not bundle a route with its conclusion**, or a firing screen will be read as a refutation of the target.

What is *not* touched. Warnaar's actual question — find a statistic stat on cylindric tableaux with $c_\alpha(t) = \sum_T t^{\mathrm{stat}(T)}$ — is untouched. What this session changes about it is the quality of the constraints:

- There are now *two proved* evaluation points, $t = 0$ and $t = 1$, not one proved and one measured.

- The two points count *different kinds of object* (Corollary 13): unbounded/ordinary with one sign class at $t = 0$, bounded/cylindric with two at $t = 1$. Any candidate statistic must therefore reconcile two different *domains*, not merely two values; in particular at $t = 0$ the count is not a count over the cylindric fibre at all.

- By Theorem 14 the two domains coincide precisely when $n \leq w$, so every fibre with $n \leq 2\ell + 2$ is useless for distinguishing candidates and every test must be run at $n > 2\ell + 2$. This retires a class of vacuous test domains — including four of the eight on which the $t = 0$ hypothesis was originally measured.

The brief's Arm 2 is retired as a route to Q442, and is not retired as a question. Arm 2 proposed to insert a formal z into HKKO's Proposition 6.1 (`prop:general`, source line 2143) and ask whether their Ishikawa–Wakayama minor summation (`thm:IsWa`, source line 1443) is winding-graded. I did not run it, and Remark 4 says why: the gap it was built to close is not on the route to $c_\alpha(0)$. Independently of Q442, the question of whether HKKO's machinery is graded is live and is *already allocated* — it is **Q444** ("relax HKKO Prop 6.1(ii) from $Q^\vee$-invariance to $Q^\vee$-equivariance, or grade by the $\mathbb{Z}/2$-orbits $\{\kappa, -\kappa\}$"), opened by this morning's browse session. So no new question number is drawn here; the allocator stays at 447. What this session contributes to Q444 is a *negative* datum worth having: a z-grading of κ cannot be read off Warnaar's t, because the two indices are on opposite sides of (4). Whatever grading Prop 6.1 admits, it is not Warnaar's.

One honest residue. I have not characterised $\mathcal{K}_0$ intrinsically. Theorem 6 expresses it as $\mathcal{Y}_0$ minus two explicit Schur sums, which is enough to refute sector matching, but it leaves open whether $\mathcal{K}_0$ is the specialisation or sector of *anything* natural. Given Remark 4 I expect not — it is a truncation of one side of an identity in a variable the other side does not carry — but I have no proof of that, and "I expect not" is not a result.

References

[1] J. Huh, J. S. Kim, C. Krattenthaler, S. Okada, *Bounded Littlewood identities for cylindric Schur functions*, arXiv:2301.13117. Read at `deep-read` level from the authors' LaTeX source (252,372 B, 6,944 lines) at `library/sources/20261009pm-cit/warnaar-cite_HKKO-2301.13117.tex`; locators in this paper are source line numbers of that file.

[2] S. O. Warnaar, *Jacobi–Trudi formulas and q,t-Rogers–Ramanujan identities*, arXiv:2511.17034 (SIGMA 26 (2026), 062). Read at `deep-read` level from source (183,950 B) at `library/sources/20261009pm-cit/warnaar-2511.17034-sigma26-062.tex`.

[3] J. R. Stembridge, *Nonintersecting paths, Pfaffians, and plane partitions*, Adv. Math. **83** (1990), 96–131. Cited only through [1]'s attribution of ([1, Eq. (1.8), source line 1169]); **not** on disk, and **not** relied upon — see Remark 12.